

AS Chemistry homework · 3.3 Metallic bonding

(and covalent / dative / dot-and-cross)

CIE 9701 AS Chemistry · 19 marks · about 40 minutes

Topic: 3.3 metallic bonding; 3.4 LO1 covalent and coordinate (dative) bonding, including Al_2Cl_6 and NH_4^+ ; 3.7 dot-and-cross. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; use CIE language (electrostatic attraction, delocalised electrons, shared pair, coordinate/dative).

Section	Content	Marks
A	Multiple choice	8
B1	2025 FM Paper 22 Question 2(a)	2
B2	2024 MJ Paper 22 Question 1(c)	1
B3	2024 MJ Paper 21 Question 5(a)	1
B4	2021 MJ Paper 22 Question 2(b)	3
B5	2024 MJ Paper 23 Question 4(a)	2
B6	2021 FM Paper 22 Question 2(e)(i)	2
Total		19

Section A — Multiple-choice questions (8 marks)**A1.**

[1]

Which statement explains why calcium has a higher melting point than barium?

- A** Calcium cations are smaller than barium cations and have a stronger attraction to the delocalised electrons.
B The structure of calcium is partly giant molecular.
C There are more delocalised electrons in calcium than in barium as it has a lower ionisation energy.
D There is greater repulsion between barium atoms as they have more complete electron shells than calcium atoms.

2023 MJ Paper 11 Question 2

A2.

[1]

Which statements help to explain the increase in melting point from sodium to aluminium?

- The charge on the metal ion increases.
- There are more delocalised electrons per metal ion.
- The radius of the metal ion decreases.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

2021 MJ Paper 13 Question 36

A3.

[1]

In which species does the underlined atom have an incomplete outer shell?

- A** $\underline{\text{B}}\text{F}_3$ **B** $\text{C}\underline{\text{H}}_3^-$ **C** $\text{F}_2\underline{\text{O}}$ **D** $\text{H}_3\underline{\text{O}}^+$

2024 MJ Paper 12 Question 6

A4.

[1]

Which species can accept a lone pair of electrons to form a coordinate (dative covalent) bond?

- BF_3
- H^+
- CH_3^+

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

2021 FM Paper 12 Question 34

A5.

[1]

When solid aluminium chloride is heated, Al_2Cl_6 is formed.Which bonding is present in Al_2Cl_6 ?

- A** covalent and coordinate (dative covalent)
B covalent only
C ionic and coordinate (dative covalent)
D ionic only

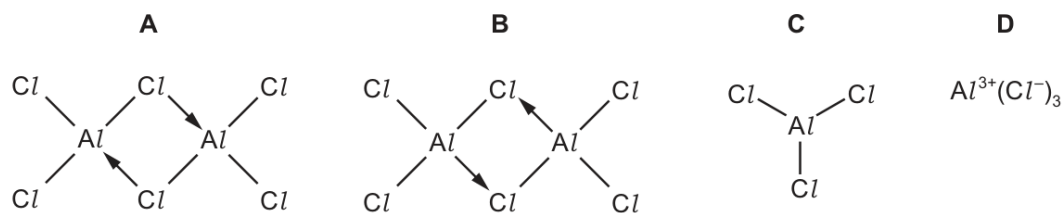
2024 MJ Paper 11 Question 5

A6.

[1]

- 5 Solid aluminium chloride sublimes at 178 °C.

Which structure best represents the species in the vapour at this temperature?

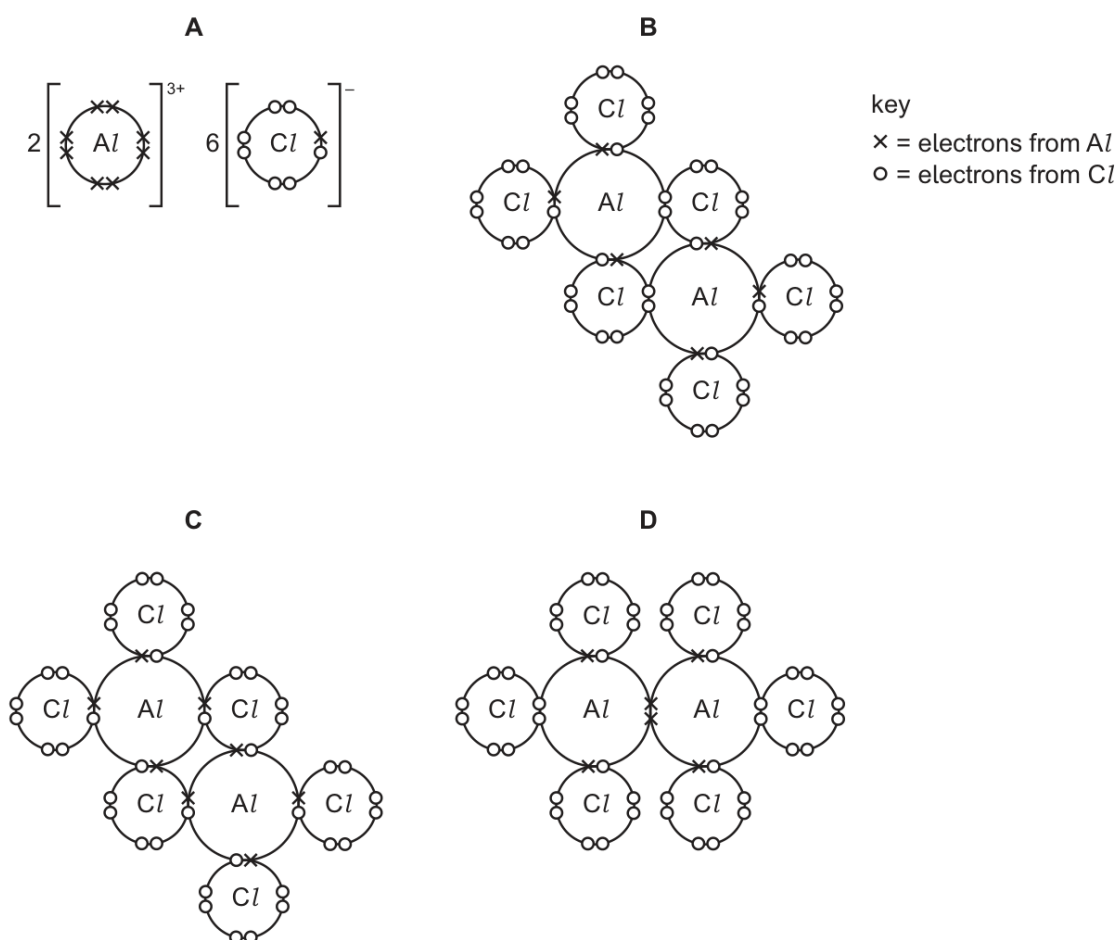


2023 ON Paper 11 Question 5

A7.

[1]

- 5 Which dot-and-cross diagram is correct for Al_2Cl_6 ?



2022 FM Paper 12 Question 5

A8.

[1]

Which elements form a chloride in which both covalent bonding and coordinate (dative covalent) bonding are present?

1. Al
2. Si
3. Mg

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

2021 ON Paper 11 Question 35

Section B — Structured questions (11 marks)**B1 · 2025 February/March Paper 22 Question 2(a) (2 marks)**

The Group 2 elements show metallic bonding.

B1(a)

[2]

Define metallic bonding.

B2 · 2024 May/June Paper 22 Question 1(c) (1 mark)**B2(c)**

[1]

Draw a labelled diagram to show the structure and bonding in sodium.

B3 · 2024 May/June Paper 21 Question 5(a) (1 mark)

Hydrocarbon molecules contain covalent bonds.

B3(a)

[1]

Define covalent bond.

B4 · 2021 May/June Paper 22 Question 2(b) (3 marks)

Carbon monoxide, CO, is a gas at room temperature and pressure. It contains a coordinate bond.

B4(b)(i)

[1]

Explain what is meant by *coordinate bond*.

B4(b)(ii)

[2]

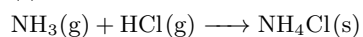
Draw a 'dot-and-cross' diagram to show the arrangement of outer electrons in CO.

Show the electrons belonging to the C atom as ×.

Show the electrons belonging to the O atom as ●.

B5 · 2024 May/June Paper 23 Question 4(a) (2 marks)

$\text{NH}_3(\text{g})$ reacts with $\text{HCl}(\text{g})$ to produce $\text{NH}_4\text{Cl}(\text{s})$, as shown.

**B5(a)**

[2]

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl .

B6 · 2021 February/March Paper 22 Question 2(e)(i) (2 marks)

Aluminium reacts with chlorine to form aluminium chloride.

Aluminium chloride can exist as the gaseous molecule $\text{Al}_2\text{Cl}_6(\text{g})$. This molecule contains coordinate bonds.

B6(e)(i)

[2]

Draw a diagram that clearly shows all the types of bond present in $\text{Al}_2\text{Cl}_6(\text{g})$.